

BACnet/SC Hub (B-SCHUB) - Fact Sheet

Chipkin Automation Systems · Version 1.5.0 · <https://github.com/chipkin/BACnetProfileExample-B-SCHUB-CPP>

A **BACnet Secure Connect hub** built on the [CAS BACnet Stack](#). BACnet/SC devices connect to it over TLS 1.3 WebSockets and it relays their BACnet traffic between them. A BACnet/IP port keeps it visible to existing BACnet/IP tools, or can be turned off for a BACnet/SC-only site.

Intended use: evaluation, interoperability testing and development of BACnet/SC devices and networks. This edition accepts **up to 4 BACnet/SC connections**. For a production hub with more connections, or to build your own product on the CAS BACnet Stack, contact support@chipkin.com.

At a glance

Item	Details
BACnet device profile	B-SCHUB - BACnet/SC Hub (ANSI/ASHRAE 135 Annex L)
BIBBs	NM-SCH-B, DS-RP-B, DS-RPM-B, DM-DDB-B, DM-DOB-B, DM-DCC-B
BACnet Protocol_Revision	30
Data links	BACnet/SC (Annex AB) hub function, plus optional hub connector with failover; BACnet/IP (Annex J), optional
Services executed	ReadProperty, ReadPropertyMultiple, Who-Is/I-Am, Who-Has/I-Have, DeviceCommunicationControl, AtomicReadFile, AtomicWriteFile and WriteProperty (certificate files), ReinitializeDevice (ACTIVATE_CHANGES, WARMSTART)
Objects	Device, Analog Input, Network Port (BACnet/IP and BACnet/SC), File (4 certificate files)
BACnet/SC connections	Up to 4 at a time (this edition)
Platforms	Windows 10/11 and Server 2016+ (x64); Linux x64 (current distributions; source build for others)
Packaging	Windows installer with a Windows service; Debian/Ubuntu .deb and Linux archive with a systemd service; signed downloads and checksums

Security

- **TLS 1.3 only**, with **mutual certificate authentication** for every BACnet/SC connection; WebSocket subprotocol `hub.bsc.bacnet.org`.
- **Certificate management over BACnet** (ANSI/ASHRAE 135 clause 19.8.3): replace the hub's certificate and add a second trusted CA remotely; new certificates are validated before they are used, so a mistake can't lock the hub out.
- **Certificate revocation** with CRLs, fail-closed.
- **Connection rate limiting** per source address and in total, before the TLS handshake; per-connection transmit limits.

- **Audit trail** naming each device by address, certificate, VMAC and device UUID.
- **BACnet/SC-only mode** for sites that allow no unencrypted BACnet traffic.
- **Certificate set-up guide** in the browser: sign a device's CSR (or a CARI request) or make its files, see every field of every file, and see live why a connection attempt failed and what to change. Signing is gated by the hub key's password.
- **CARI files** (ANSI/ASHRAE 135-2024 Annex AA.2) in and out, for BACCARI and other CA tools; lab certificate generator for testing; guide for production PKI.

Operations

- Status page, /health (for monitors and load balancers) and /metrics over HTTP or HTTPS.
- Rotating log files; configuration file; per-site device name and network numbers.
- Software bill of materials (CycloneDX) with every release.

Documents

- **User manual** - manual.pdf
- **PICS** (Protocol Implementation Conformance Statement) - PICS.pdf
- **Releases** - <https://github.com/chipkin/BACnetProfileExample-B-SCHUB-CPP/releases>
- **CAS BACnet Stack** - <https://store.chipkin.com/services/stacks/bacnet-stack>

Contact: Chipkin Automation Systems · support@chipkin.com · <https://store.chipkin.com/contact-us>